

Qualification Pack



Parali Brick Fuel Machine Operator

QP Code: RSC/Q8701

Version: 1.0

NSQF Level: 2

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RSC/Q8701: Parali Brick Fuel Machine Operator

Brief Job Description

Parali Brick Fuel Machine Operator is responsible for assisting in the process of converting agricultural stubble (parali) into briquettes. The role involves operating and maintaining stubble briquetting machine, handling chemicals, binders, and nanomaterials (as per requirement) and ensuring the quality of the final product. The Parali Brick Fuel Machine Operator will work under the supervision of senior operators and follow safety protocols to ensure a safe and efficient production environment. This role plays a crucial part in reducing stubble burning and promoting energy practices.

Personal Attributes

The candidate should demonstrate strong attention to detail, a commitment to safety, and the ability to work collaboratively in a team-oriented environment. They should also possess problem-solving skills and a proactive attitude towards maintenance and quality assurance in the briquetting process.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [RSC/N0807: Operate Biomass/ Stubble Briquetting Machinery](#)
2. [RSC/N8801: Handle Chemical and Binders Safely](#)
3. [RSC/N8302: Perform routine maintenance and quality checks](#)
4. [RSC/N5007: Follow health & safety protocol](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Rubber
Sub-Sector	Chemical & Petrochemical
Occupation	Waste Management, Storage and Handling, Chemical Application, Manufacturing/process
Country	India
NSQF Level	2

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Credits	11
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 8131
Minimum Educational Qualification & Experience	Ability to read and write
Minimum Level of Education for Training in School	Ability to read and write
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	22/10/2027
NSQC Approval Date	22/10/2024
Version	1.0
Reference code on NQR	QG-02-RI-03326-2024-V1-RCPSDC
NQR Version	1

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RSC/N0807: Operate Biomass/ Stubble Briquetting Machinery

Description

This unit is about preparing stubble and machinery for briquetting operation and operating briquetting machine to produce briquettes.

Scope

The scope covers the following :

- Prepare Stubble for Briquetting Operation
- Handling and Preparation of Stubble/ Biomass
- Operate Briquetting Machine

Elements and Performance Criteria

Prepare Stubble for Briquetting Operation

To be competent, the user/individual on the job must be able to:

- PC1.** inspect and select stubble suitable for briquetting, ensuring that the material is free from unwanted debris and contaminants
- PC2.** measure moisture content of stubble using appropriate tools and techniques to ensure it meets briquetting standards
- PC3.** prepare stubble by cutting it to the required size and shape using appropriate cutting tools or machines
- PC4.** set up and calibrate pre-treatment machinery (if applicable) to process stubble effectively
- PC5.** follow environmental and safety guidelines during the handling and preparation of stubble to avoid fire hazards

Handling and Preparation of Stubble Biomass

To be competent, the user/individual on the job must be able to:

- PC6.** load stubble into storage areas or conveyors for transportation to the briquetting area
- PC7.** mix stubble biomass with required binders, chemicals, or nanomaterials (as per the product specification)
- PC8.** check the quality and consistency of the mixture to ensure uniformity for briquette formation
- PC9.** operate auxiliary equipment for handling biomass, such as conveyors, balers, or shredders, as required
- PC10.** maintain cleanliness of the work area and handle chemicals or binders following safety procedures

Operate Briquetting Machine

To be competent, the user/individual on the job must be able to:

- PC11.** start, operate, and monitor the briquetting machine following standard operating procedures (SOPs)
- PC12.** adjust machine settings to ensure briquettes meet the required dimensions, density, and quality standards

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- PC13.** continuously monitor the flow of biomass into the machine to ensure smooth operation and avoid downtime
- PC14.** identify and troubleshoot minor issues with the machine, such as blockages, overheating, or wear and tear
- PC15.** safely shut down the machine and perform basic maintenance, such as cleaning filters and lubricating parts, as per the maintenance schedule
- PC16.** record production data and report any defects or malfunctions to the supervisor promptly
- PC17.** follow all safety protocols, including the use of personal protective equipment (PPE) while operating machinery

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the environmental and economic implications of stubble burning and its contribution to air pollution
- KU2.** benefits of converting stubble into briquettes for sustainable energy production
- KU3.** fundamentals of the briquetting process and its role in renewable energy generation
- KU4.** different types of briquetting machines (hydraulic, screw, piston) and their basic working principles
- KU5.** types of stubble and biomass suitable for briquetting (moisture content, size, and quality)
- KU6.** importance of binders, chemicals, and nanomaterials in enhancing the quality of briquettes
- KU7.** the proportion and mixing techniques for stubble and binders/chemicals
- KU8.** the operation and maintenance of the briquetting machine, including standard operating procedures (SOPs)
- KU9.** basic troubleshooting techniques for common machine issues
- KU10.** safety standards and the correct use of personal protective equipment (PPE)
- KU11.** environmental regulations concerning biomass handling, chemical use, and briquette production
- KU12.** guidelines for waste management and recycling in the context of briquetting operations
- KU13.** familiarity with key components of a briquetting machine such as the press chamber, hydraulic systems, sensors, and control panels
- KU14.** about auxiliary equipment like conveyors, shredders, and pre-treatment machines
- KU15.** quality parameters such as briquette density, size, shape, and moisture content
- KU16.** techniques for sampling and testing briquettes to ensure consistency with product specifications
- KU17.** techniques for pre-processing biomass, including cutting, drying, and mixing
- KU18.** about tools used for measuring moisture content and particle size
- KU19.** preventive maintenance procedures, such as lubricating, cleaning, and replacing machine parts
- KU20.** common issues like machine blockages, overheating, and mechanical wear and tear
- KU21.** use of logbooks or digital systems to record production data, equipment usage, and machine performance

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KU22. process of reporting defects or issues to supervisors

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and comprehend instructions, standard operating procedures (SOPs), and safety guidelines related to briquetting operations
- GS2.** maintain logs, filling out production reports, and recording machine maintenance data accurately
- GS3.** effectively communicate with supervisors, senior operators, and team members regarding machine operations, production progress, and any arising issues
- GS4.** verbal reporting of malfunctions, safety concerns, or maintenance needs
- GS5.** engage in discussions during team briefings or safety meetings to clarify instructions and resolve doubts
- GS6.** make quick decisions during machine operations, such as adjusting settings or halting the machine in case of an emergency or defect detection
- GS7.** take independent action in minor troubleshooting situations and knowing when to escalate issues to supervisors
- GS8.** plan and organize daily tasks to ensure continuous and efficient operation of the briquetting machine
- GS9.** prioritize tasks such as machine preparation, material handling, and regular machine maintenance to meet production targets and avoid downtime
- GS10.** identify and solve operational issues, such as material jams, variations in briquette quality, or equipment malfunction
- GS11.** optimize machine performance and ensuring smooth material flow during production
- GS12.** work collaboratively with other operators, technicians, and supervisors to ensure smooth workflow and efficient production
- GS13.** offer assistance to peers and sharing knowledge related to machine operation or stubble handling techniques
- GS14.** manage time effectively to complete machine setup, material handling, and production tasks within the allocated shift hours
- GS15.** ensure timely communication and reporting to supervisors to avoid production delays

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare Stubble for Briquetting Operation</i>	9	20	-	-
PC1. inspect and select stubble suitable for briquetting, ensuring that the material is free from unwanted debris and contaminants	2	4	-	-
PC2. measure moisture content of stubble using appropriate tools and techniques to ensure it meets briquetting standards	2	4	-	-
PC3. prepare stubble by cutting it to the required size and shape using appropriate cutting tools or machines	2	4	-	-
PC4. set up and calibrate pre-treatment machinery (if applicable) to process stubble effectively	2	4	-	-
PC5. follow environmental and safety guidelines during the handling and preparation of stubble to avoid fire hazards	1	4	-	-
<i>Handling and Preparation of Stubble Biomass</i>	7	20	-	-
PC6. load stubble into storage areas or conveyors for transportation to the briquetting area	1	4	-	-
PC7. mix stubble biomass with required binders, chemicals, or nanomaterials (as per the product specification)	2	4	-	-
PC8. check the quality and consistency of the mixture to ensure uniformity for briquette formation	1	4	-	-
PC9. operate auxiliary equipment for handling biomass, such as conveyors, balers, or shredders, as required	2	4	-	-
PC10. maintain cleanliness of the work area and handle chemicals or binders following safety procedures	1	4	-	-
<i>Operate Briquetting Machine</i>	14	30	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. start, operate, and monitor the briquetting machine following standard operating procedures (SOPs)	2	5	-	-
PC12. adjust machine settings to ensure briquettes meet the required dimensions, density, and quality standards	2	5	-	-
PC13. continuously monitor the flow of biomass into the machine to ensure smooth operation and avoid downtime	2	4	-	-
PC14. identify and troubleshoot minor issues with the machine, such as blockages, overheating, or wear and tear	2	4	-	-
PC15. safely shut down the machine and perform basic maintenance, such as cleaning filters and lubricating parts, as per the maintenance schedule	2	4	-	-
PC16. record production data and report any defects or malfunctions to the supervisor promptly	2	4	-	-
PC17. follow all safety protocols, including the use of personal protective equipment (PPE) while operating machinery	2	4	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N0807
NOS Name	Operate Biomass/ Stubble Briquetting Machinery
Sector	Rubber
Sub-Sector	Natural Rubber (NR) Plantation, Non - Tyre
Occupation	Production, Storage and Handling
NSQF Level	2
Credits	4
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024

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RSC/N8801: Handle Chemical and Binders Safely

Description

This unit is about handling chemicals and binders and applying it on parali before/ during briquetting process.

Scope

The scope covers the following :

- Handling Binders
- Handling Chemicals
- Applying Chemicals and Binders on Parali

Elements and Performance Criteria

Handling Binders

To be competent, the user/individual on the job must be able to:

- PC1.** identify and select appropriate binders as per the production requirements and quality standards
- PC2.** measure and prepare binders in the correct proportions, ensuring accuracy according to product specifications
- PC3.** safely store binders in designated areas, ensuring containers are properly sealed and labeled to avoid contamination
- PC4.** handle binders using appropriate personal protective equipment (PPE) such as gloves, masks, and goggles to avoid direct contact
- PC5.** maintain cleanliness in the binder handling area to prevent spills or cross-contamination with other materials

Handling Chemicals

To be competent, the user/individual on the job must be able to:

- PC6.** identify and select chemicals used in the briquetting process, including additives or preservatives, as per standard procedures
- PC7.** measure chemicals accurately using suitable measuring tools to avoid incorrect dosage and ensure uniform application
- PC8.** store chemicals in a safe and secure location, following the organization's safety protocols for labeling, storage temperature, and hazard classification
- PC9.** handle chemicals while following all safety regulations, including the use of PPE, proper ventilation, and safe handling techniques to avoid inhalation or skin contact
- PC10.** follow spill management procedures in case of accidental chemical spills and immediately report to the supervisor for further action

Applying Chemicals and Binders on Parali

To be competent, the user/individual on the job must be able to:

- PC11.** mix binders and chemicals with the stubble biomass in the correct ratio as per the technical guidelines for briquetting

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- PC12.** ensure even and uniform application of chemicals and binders on the parali using the appropriate mixing equipment or techniques
- PC13.** monitor the consistency and quality of the chemical-binder mixture during the application to ensure that it meets the required standards for briquette formation
- PC14.** follow safety procedures while applying chemicals and binders to avoid exposure and contamination, ensuring all equipment is cleaned after use
- PC15.** record the quantities and types of chemicals and binders applied, along with any observations related to the quality of the mixture, in the production log

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** various binders (organic, inorganic) and chemicals used in the stubble briquetting process
- KU2.** how binders enhance the briquette structure and the role of chemicals in improving briquette durability and efficiency
- KU3.** safety standards, government regulations, and guidelines for handling hazardous chemicals and materials
- KU4.** physical and chemical properties such as toxicity, flammability, and environmental hazards of the materials handled
- KU5.** the environmental consequences of improper handling and disposal of binders and chemicals, and the importance of eco-friendly practices
- KU6.** measuring tools and techniques for accurately mixing chemicals and binders
- KU7.** correct storage methods for chemicals and binders, including temperature control, labeling, and segregation
- KU8.** the methods and tools used for applying chemicals and binders evenly on stubble biomass
- KU9.** the appropriate personal protective equipment (PPE) required when handling chemicals and binders
- KU10.** first-aid procedures, spill management, and emergency responses for chemical accidents
- KU11.** the organization's safety policies regarding chemical handling, including the use of PPE, hazard communication, and incident reporting
- KU12.** workplace rules and regulations related to chemical storage, handling, and disposal, as per organizational and legal standards

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read safety instructions, chemical labels, material safety data sheets (MSDS), and standard operating procedures (SOPs)
- GS2.** document chemical usage, quantities of binders, and any incidents or deviations from standard processes
- GS3.** clearly communicate with team members and supervisors about safety issues, material handling, and any operational challenges

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- GS4.** engage in discussions during safety briefings and team meetings to ensure mutual understanding of tasks and safety protocols
- GS5.** make quick and sound decisions in case of chemical spills, leaks, or accidents, ensuring the safety of self and others
- GS6.** use PPE and other safety measures during chemical handling based on risk assessment
- GS7.** plan the sequence of tasks, including preparing chemicals, applying binders, and cleaning equipment, to maintain smooth production flow
- GS8.** ensure compliance with safety guidelines and maintaining chemical inventories
- GS9.** address issues such as improper chemical mixtures or equipment malfunctions and finding practical solutions to maintain production efficiency
- GS10.** identify risks in chemical handling and mitigating them to prevent accidents
- GS11.** evaluate the effectiveness of chemical and binder mixtures by analyzing the quality of briquettes and adjusting the process accordingly
- GS12.** assess safety risks during chemical handling and making informed decisions to minimize exposure and hazards
- GS13.** collaborate effectively with other operators and supervisors to ensure the safe and efficient handling of chemicals and binders
- GS14.** follow safety protocols to prevent exposure to hazardous substances

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Handling Binders</i>	10	24	-	-
PC1. identify and select appropriate binders as per the production requirements and quality standards	2	4	-	-
PC2. measure and prepare binders in the correct proportions, ensuring accuracy according to product specifications	2	6	-	-
PC3. safely store binders in designated areas, ensuring containers are properly sealed and labeled to avoid contamination	2	4	-	-
PC4. handle binders using appropriate personal protective equipment (PPE) such as gloves, masks, and goggles to avoid direct contact	2	6	-	-
PC5. maintain cleanliness in the binder handling area to prevent spills or cross-contamination with other materials	2	4	-	-
<i>Handling Chemicals</i>	10	22	-	-
PC6. identify and select chemicals used in the briquetting process, including additives or preservatives, as per standard procedures	2	6	-	-
PC7. measure chemicals accurately using suitable measuring tools to avoid incorrect dosage and ensure uniform application	2	4	-	-
PC8. store chemicals in a safe and secure location, following the organization's safety protocols for labeling, storage temperature, and hazard classification	2	4	-	-
PC9. handle chemicals while following all safety regulations, including the use of PPE, proper ventilation, and safe handling techniques to avoid inhalation or skin contact	2	4	-	-
PC10. follow spill management procedures in case of accidental chemical spills and immediately report to the supervisor for further action	2	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Applying Chemicals and Binders on Parali</i>	10	24	-	-
PC11. mix binders and chemicals with the stubble biomass in the correct ratio as per the technical guidelines for briquetting	2	6	-	-
PC12. ensure even and uniform application of chemicals and binders on the parali using the appropriate mixing equipment or techniques	2	6	-	-
PC13. monitor the consistency and quality of the chemical-binder mixture during the application to ensure that it meets the required standards for briquette formation	2	4	-	-
PC14. follow safety procedures while applying chemicals and binders to avoid exposure and contamination, ensuring all equipment is cleaned after use	2	4	-	-
PC15. record the quantities and types of chemicals and binders applied, along with any observations related to the quality of the mixture, in the production log	2	4	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N8801
NOS Name	Handle Chemical and Binders Safely
Sector	Rubber
Sub-Sector	Natural Rubber (NR) Plantation, Non - Tyre
Occupation	Chemical Application, Storage and Handling, Storage and Handling
NSQF Level	2
Credits	3
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024

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RSC/N8302: Perform routine maintenance and quality checks

Description

This unit is about regularly inspecting the machinery and carrying out routine maintenance as required. Also includes performing quality checks on manufactured briquettes.

Scope

The scope covers the following :

- Regular Inspection of Machinery
- Carry Out Maintenance of Machinery
- Carry Out Quality Checks on Manufactured Briquettes

Elements and Performance Criteria

Regular Inspection of Machinery

To be competent, the user/individual on the job must be able to:

- PC1.** conduct daily visual inspections of the briquetting machine for signs of wear, damage, or malfunction before starting the operation
- PC2.** check all key components, including rollers, press plates, and feeding mechanisms, for any operational irregularities
- PC3.** ensure that safety guards and emergency stops are functional and not tampered with
- PC4.** inspect electrical connections and hydraulic systems for leaks, loose wires, or potential hazards
- PC5.** report any irregularities or mechanical issues to senior operators or supervisors for further action

Carry Out Maintenance of Machinery

To be competent, the user/individual on the job must be able to:

- PC6.** lubricate moving parts of the briquetting machine as per the manufacturer's recommendations to reduce wear and tear
- PC7.** tighten loose bolts, nuts, or fittings to maintain machine stability and operational efficiency
- PC8.** replace or clean filters, belts, and any worn-out parts of the machine according to the maintenance schedule
- PC9.** maintain a logbook of maintenance activities, recording the date, time, and type of service performed on the machinery
- PC10.** ensure the machine is cleaned and free from stubble residues or debris after each operation to prevent clogging and maintain efficiency

Carry Out Quality Checks on Manufactured Briquettes

To be competent, the user/individual on the job must be able to:

- PC11.** inspect briquettes for uniformity in size, shape, and density during and after the production process
- PC12.** measure moisture content in briquettes to ensure it meets the required quality standards for fuel efficiency

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- PC13.** check for any visible defects such as cracks, inconsistencies, or brittleness in the briquettes
- PC14.** conduct random sample testing of briquettes for mechanical strength and durability
- PC15.** record observations from the quality checks and report any deviations from standard quality to the supervisor for corrective actions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the key components and working principles of the briquetting machine (e.g., rollers, press plates, feeding systems)
- KU2.** standard maintenance practices such as lubrication, tightening, and replacement of worn-out parts to ensure optimal machine performance
- KU3.** safety protocols, including how to inspect and maintain safety devices like emergency stops, guards, and electrical systems
- KU4.** how to conduct visual and mechanical inspections to identify potential machine issues (e.g., wear, damage, misalignment)
- KU5.** how to get familiar with the required quality parameters for briquettes, such as uniformity, moisture content, and mechanical strength
- KU6.** organization's preventive maintenance schedules and procedures for the briquetting machine
- KU7.** document maintenance activities, inspections, and quality check results as per organizational requirements
- KU8.** organizational protocols for managing spare parts, tools, and materials required for routine machine maintenance
- KU9.** how to report equipment issues, safety concerns, or quality deviations to supervisors and team members
- KU10.** the role of routine maintenance and quality checks in promoting sustainable production and minimizing downtime in line with the organization's green initiatives

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and understand maintenance manuals, safety guidelines, and technical specifications of the briquetting machine
- GS2.** record maintenance activities, inspection results, and quality check observations in logs or digital systems
- GS3.** effectively communicate with supervisors and team members to report maintenance needs, quality issues, and machine malfunctions
- GS4.** engage in discussion during team meetings to ensure everyone is aligned on machine upkeep and quality standards
- GS5.** make informed decisions during inspections and maintenance, such as identifying when a machine part requires replacement or immediate attention

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- GS6.** organize maintenance tasks in a timely manner to ensure minimal disruption in production schedules while maintaining machine efficiency
- GS7.** troubleshoot minor machine issues, such as detecting mechanical faults or quality defects, and take corrective actions or report to senior operators for resolution
- GS8.** analyze inspection results and quality checks to assess machine performance and make recommendations for adjustments or improvements
- GS9.** evaluate the root cause of recurring maintenance problems and developing preventive measures to avoid equipment downtime
- GS10.** carry out maintenance tasks, inspecting machine components, and ensuring quality consistency in briquette production
- GS11.** work collaboratively with other operators and maintenance personnel to ensure smooth machine operations and quality control
- GS12.** understand the importance of product quality and machine efficiency in meeting customer demands for high-quality briquettes, contributing to the organization's reputation and sustainability goals

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Regular Inspection of Machinery</i>	10	20	-	-
PC1. conduct daily visual inspections of the briquetting machine for signs of wear, damage, or malfunction before starting the operation	2	4	-	-
PC2. check all key components, including rollers, press plates, and feeding mechanisms, for any operational irregularities	2	4	-	-
PC3. ensure that safety guards and emergency stops are functional and not tampered with	2	4	-	-
PC4. inspect electrical connections and hydraulic systems for leaks, loose wires, or potential hazards	2	4	-	-
PC5. report any irregularities or mechanical issues to senior operators or supervisors for further action	2	4	-	-
<i>Carry Out Maintenance of Machinery</i>	10	24	-	-
PC6. lubricate moving parts of the briquetting machine as per the manufacturer's recommendations to reduce wear and tear	2	5	-	-
PC7. tighten loose bolts, nuts, or fittings to maintain machine stability and operational efficiency	2	5	-	-
PC8. replace or clean filters, belts, and any worn-out parts of the machine according to the maintenance schedule	2	5	-	-
PC9. maintain a logbook of maintenance activities, recording the date, time, and type of service performed on the machinery	2	4	-	-
PC10. ensure the machine is cleaned and free from stubble residues or debris after each operation to prevent clogging and maintain efficiency	2	5	-	-
<i>Carry Out Quality Checks on Manufactured Briquettes</i>	10	26	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. inspect briquettes for uniformity in size, shape, and density during and after the production process	2	5	-	-
PC12. measure moisture content in briquettes to ensure it meets the required quality standards for fuel efficiency	2	6	-	-
PC13. check for any visible defects such as cracks, inconsistencies, or brittleness in the briquettes	2	5	-	-
PC14. conduct random sample testing of briquettes for mechanical strength and durability	2	5	-	-
PC15. record observations from the quality checks and report any deviations from standard quality to the supervisor for corrective actions	2	5	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N8302
NOS Name	Perform routine maintenance and quality checks
Sector	Rubber
Sub-Sector	Natural Rubber (NR) Plantation, Non - Tyre
Occupation	Maintenance, Storage and Handling, Storage and Handling, Quality Control
NSQF Level	2
Credits	2
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024

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RSC/N5007: Follow health & safety protocol

Description

This unit is about maintaining health and safety of self and others at workplace.

Scope

The scope covers the following :

- Maintain a clean and efficient workplace
- Render appropriate emergency procedures
- Maintain standard safety procedures at the workplace
- Participate in safety awareness campaigns
- Understand potential sources of accidents
- Use safety gears to avoid accidents

Elements and Performance Criteria

Maintain a clean and efficient workplace

To be competent, the user/individual on the job must be able to:

- PC1.** undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor
- PC2.** work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy
- PC3.** read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc
- PC4.** prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.
- PC5.** use equipment and materials safely and correctly and return the same to designated storage when not in use
- PC6.** dispose off waste safely and correctly in a designated area
- PC7.** risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace
- PC8.** perform work in a manner which minimizes environmental damage
- PC9.** all procedures and work instructions for controlling risk are followed closely
- PC10.** report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger

Render appropriate emergency procedures

To be competent, the user/individual on the job must be able to:

- PC11.** follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.
- PC12.** follow emergency procedures as per company standards and workplace requirements

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- PC13.** use emergency equipment in accordance with manufacturers' specifications and workplace requirements
- PC14.** provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.
- PC15.** recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate
- PC16.** dispose off medical waste in accordance with workplace requirements
- PC17.** report details of first aid administered in accordance with work place procedures

Maintain standard safety procedures at the workplace

To be competent, the user/individual on the job must be able to:

- PC18.** comply with general safety procedures
- PC19.** follow standard safety procedures while handling equipment, hazardous material or tool
- PC20.** check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.
- PC21.** ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure
- PC22.** keep the workplace organized, swept, clean and hazard free

Participate in safety awareness campaigns

To be competent, the user/individual on the job must be able to:

- PC23.** attend fire drills and other safety related workshops organized at the workplace
- PC24.** be aware of first aid, evacuation and emergency procedures
- PC25.** be alert of any events and do not be negligent to any safety procedures to be followed

Understand potential sources of accidents

To be competent, the user/individual on the job must be able to:

- PC26.** avoid accidents while using hazardous chemicals, machines, sharp tools and equipment

Use safety gears to avoid accidents

To be competent, the user/individual on the job must be able to:

- PC27.** use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)
- PC28.** handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies on incentives, delivery standards, and personnel management.
- KU2.** occupational safety and health policy followed
- KU3.** emergency evacuation procedure
- KU4.** medical policy
- KU5.** company laws and acts
- KU6.** the risks to health and safety and the measures to be taken to control those risks in the area of work

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- KU7.** workplace procedures and requirements for the handling of workplace injuries/illnesses.
- KU8.** basic emergency first aid procedure
- KU9.** local emergency services
- KU10.** reporting on accidents, incidents and problems to appropriate authorities.
- KU11.** how to use machines as per standard operating procedure
- KU12.** how to maintain work area safe and secure
- KU13.** use of hazardous materials, tools and equipments
- KU14.** emergency evacuation and first aid procedures to be followed
- KU15.** personal hygiene and fitness requirements
- KU16.** general duties under the relevant health and safety legislation
- KU17.** what personal protective equipment and clothing should be worn and how it is cared for
- KU18.** the correct and safe way to use materials and equipment required for work
- KU19.** the importance of good housekeeping in the workplace
- KU20.** safe disposal methods for waste
- KU21.** methods for minimizing environmental damage during work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data which are required for record keeping purpose
- GS2.** report problems to the appropriate person in a timely manner
- GS3.** write descriptions and details about incidents in reports
- GS4.** read instruction manuals for hand tools and equipment
- GS5.** read instructions on work orders and procedures
- GS6.** receive instructions and seek advice from superiors
- GS7.** communicate clearly and effectively with others
- GS8.** take a decision for any change/issue based on earlier successes(documented previous history)on similar issues
- GS9.** work out changes in case a new improved machine/equipment is added in the process or any new material/chemical is developed replacing existing one.
- GS10.** make changes in cycle time due to improved process.
- GS11.** use the standard operating procedure or trouble shooting manuals for trouble shooting and other reference documents approved by plant management
- GS12.** consult the peer group and superiors to arrive at a favourable decision.
- GS13.** use of standard available problem solving techniques for decision making
- GS14.** review and analyze the process steps to check on system non adherence andnon conformity
- GS15.** review the current sop and other standards for continuous improvement to facilitate decision making
- GS16.** take a calculated risk with minimum losses
- GS17.** schedule daily activities and drawing up priorities; allocate start times, estimation of completion times and materials, equipment and assistance required for completion.

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- GS18.** match customer needs/specification by adjusting the processing conditions (interact with customer in case any clarification required)
- GS19.** ensure that performance of his action/operation/activity does not lead to any divergence from the specified quality of the final product as required by the customer.
- GS20.** complete the assigned task in timely manner so that the final product is delivered in the timeline given by the customer.
- GS21.** communicate effectively to the superior/customer for any delay in supplies to the clients.
- GS22.** work towards fulfilling the customers requirement as per their demand.
- GS23.** in case of any complaint, ensure its timely resolution if the problem is emanating at his level
- GS24.** communicate effectively to the superior/customer for any delay in resolving the problem faced by the customer.
- GS25.** maintain good/cordial relation with customers.
- GS26.** work on the feedback received from customer regarding the product.
- GS27.** use first aid treatment in case of any injury/accident.
- GS28.** monitor and maintain the condition of tools and equipment
- GS29.** assess situation & identify appropriate control measures
- GS30.** act, communicate and report in emergency situation

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain a clean and efficient workplace</i>	9	21	-	-
PC1. undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor	2	2	-	-
PC2. work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy	2	2	-	-
PC3. read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc	-	2	-	-
PC4. prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.	2	2	-	-
PC5. use equipment and materials safely and correctly and return the same to designated storage when not in use	1	2	-	-
PC6. dispose off waste safely and correctly in a designated area	2	4	-	-
PC7. risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace	-	2	-	-
PC8. perform work in a manner which minimizes environmental damage	-	2	-	-
PC9. all procedures and work instructions for controlling risk are followed closely	-	1	-	-
PC10. report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger	-	2	-	-
<i>Render appropriate emergency procedures</i>	9	18	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.	2	4	-	-
PC12. follow emergency procedures as per company standards and workplace requirements	2	4	-	-
PC13. use emergency equipment in accordance with manufacturers' specifications and workplace requirements	2	4	-	-
PC14. provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.	-	1	-	-
PC15. recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate	-	2	-	-
PC16. dispose off medical waste in accordance with workplace requirements	-	1	-	-
PC17. report details of first aid administered in accordance with work place procedures	3	2	-	-
<i>Maintain standard safety procedures at the workplace</i>	6	15	-	-
PC18. comply with general safety procedures	2	4	-	-
PC19. follow standard safety procedures while handling equipment, hazardous material or tool	-	2	-	-
PC20. check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.	2	4	-	-
PC21. ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure	-	1	-	-
PC22. keep the workplace organized, swept, clean and hazard free	2	4	-	-
<i>Participate in safety awareness campaigns</i>	2	8	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. attend fire drills and other safety related workshops organized at the workplace	1	3	-	-
PC24. be aware of first aid, evacuation and emergency procedures	1	3	-	-
PC25. be alert of any events and do not be negligent to any safety procedures to be followed	-	2	-	-
<i>Understand potential sources of accidents</i>	1	3	-	-
PC26. avoid accidents while using hazardous chemicals, machines, sharp tools and equipment	1	3	-	-
<i>Use safety gears to avoid accidents</i>	3	5	-	-
PC27. use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)	2	2	-	-
PC28. handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders	1	3	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N5007
NOS Name	Follow health & safety protocol
Sector	Rubber
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
RSC/N0807. Operate Biomass/ Stubble Briquetting Machinery	30	70	0	0	100	30
RSC/N8801. Handle Chemical and Binders Safely	30	70	0	0	100	30
RSC/N8302. Perform routine maintenance and quality checks	30	70	0	0	100	20
RSC/N5007. Follow health & safety protocol	30	70	-	-	100	15
DGT/VSQ/N0101. Employability Skills (30 Hours)	20	30	-	-	50	5
Total	140	310	-	-	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.